# CSCI 3130 — HW4: Concise Solutions

**Q1.** $B, C$ are non-generating: every derivation from either retains a $B$ or $C$. Delete them and $S \to aBB$, leaving

$$S \to aaA, \qquad A \to aA \mid \lambda.$$

Eliminating $A \to \lambda$ gives

$$\boxed{S \to aaA \mid aa, \qquad A \to aA \mid a.}$$

There are no unit-, useless-, or $\lambda$-productions. Since originally $A$ generates $a^*$, $\boxed{L = \{a^n : n \geq 2\}}$.

**Q2.** No $\lambda$- or unit-productions occur; all variables are generating and reachable.

Introduce $X \to a$ and $Y \to b$. Replacing terminals in longer rules gives

$$A \to XAB \mid b, \qquad B \to YB \mid YAC \quad (S, C \text{ unchanged}).$$

Introduce $D \to AB$ and $E \to AC$ to split the length-three rules. The final CNF grammar is

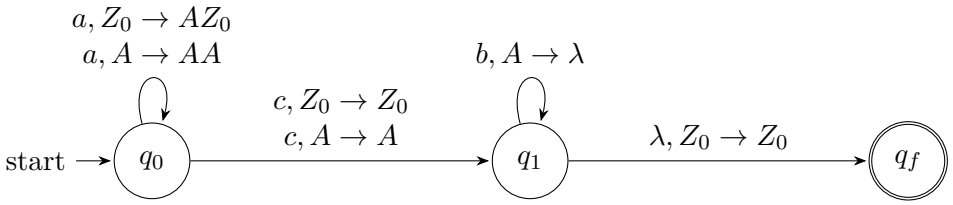
$$\boxed{\begin{array}{ll} S \to AB \mid a, & C \to c, \\ A \to XD \mid b, & D \to AB, \\ B \to YB \mid YE, & E \to AC, \\ X \to a, & Y \to b. \end{array}}$$

Every rule has the form $U \to VW$ or $U \to t$.

**Q3. (1) Deterministic.** Each transition set is a singleton. The $a$-move requires stack top $z$, whereas the $\lambda$-move requires $A$; they cannot be enabled together.

**(2) Nondeterministic.** At $(q_0, z)$ with next input $a$, both $\delta(q_0, a, z)$ and $\delta(q_0, \lambda, z)$ are enabled, violating DPDA determinism.

**Q4.** Use $Q = \{q_0, q_1, q_f\}$, $\Sigma = \{a, b, c\}$, $\Gamma = \{A, Z_0\}$. Start in $q_0$ with stack $Z_0$; $F = \{q_f\}$. Labels $x, X \to \gamma$ mean read $x$ and replace top $X$ by $\gamma$ (leftmost symbol on top). All unshown transitions are empty.




Each $a$ pushes one $A$; after the unique $c$, each $b$ pops one $A$. Acceptance requires reaching $q_f$ with all input consumed, hence equal numbers of $a$'s and $b$'s. For $n = 0$, $c$ is also accepted. Thus $L(M) = \{a^n c b^n : n \geq 0\}$.